Psychological Mechanisms for Eliciting Preferences and Beliefs*

Evan Friedman† Suanna Oh‡ Duncan Webb§

4th April 2026

Abstract

We study two psychology-based interventions for improving survey response quality: (i) an unexpected bonus payment designed to trigger reciprocity towards the researcher and (ii) telling respondents that they will later be paid for accurately restating their previously-given answers, which could motivate careful initial answers. In an online experiment, the bonus method improves both answer correctness and consistency, driven by increased effort and reciprocity. The restatement method, however, only improves consistency, as some participants channel effort toward memorizing their answers rather than answering carefully. These results demonstrate both the potential and limitations of using psychological mechanisms to improve survey response quality.

Keywords: restatement method; bonus method; incentives; survey

JEL Classification: C81, C83, C91, D91

*We thank Esther Duflo, Armin Falk, Nicolas Jacquemet, George Loewenstein, Karen Macours, Frank Schilbach, Egon Tripodi, Florian Zimmermann, and seminar participants at the Berkeley PEP meeting, MIT Behavioral Lunch, Belief-based Utility Conference 2022, briq Beliefs Workshop 2022, NHH FAIR, Erasmus School of Economics, Berlin Behavioral Economics Seminar, University of Fribourg, Aarhus BSS, University of Copenhagen, Paris Behavioral Symposium 2024, CESifo Area Conference on Behavioral Economics 2024, and ASSA 2025 Annual Meeting for helpful comments. We gratefully acknowledge financial support from the Collaborative Research Center (CRC) and Ecole Universitaire de Recherche - PGSE. Duncan Webb acknowledges support by Fundação para a Ciência e a Tecnologia (UIDB/00124/2020, UIDP/00124/2020, UID/00124, Nova School of Business and Economics and Social Sciences DataLab - PINFRA/22209/2016), and POR Lisboa and POR Norte (Social Sciences DataLab, PINFRA/22209/2016). This study was pre-registered on *AsPredicted* under the IDs #119446 (available here) and #194085 (available here).

†Paris School of Economics and CESifo, evan.friedman@psemail.eu.

‡Paris School of Economics, BREAD, CEPR, and CESifo. suanna.oh@psemail.eu

§Princeton University, dmbwebb@gmail.com

1 Introduction

Accurately measuring preferences and beliefs is a central challenge in conducting social science research. Such measurements are crucial for testing theories, estimating structural relationships, and interpreting public opinion. Yet their reliability depends on how much effort respondents invest. A typical study can expect 8-12% of respondents to answer carelessly or inattentively (Curran, 2016), which can generate systematic measurement error (Meade and Craig, 2012; Maniaci and Rogge, 2014; Meyer et al., 2015). This problem has become particularly acute with the rise of online surveys and experiments, where there are widespread concerns about data quality (Chmielewski and Kucker, 2020).

For questions whose responses can be *verified* against the ground truth, there are several standard ways to incentivize effort. We can pay respondents for choosing objectively correct answers or stating more accurate beliefs, and we can implement respondents' chosen options on simple preference questions.[1] However, while theoretically incentive-compatible, these interventions may not necessarily improve answer quality in practice (Enke et al., 2023). Furthermore, many important survey questions in social science are *unverifiable*. They may concern distant future events, subjective feelings, hypothetical scenarios, or choices over prohibitively expensive objects. This raises a fundamental question: how can researchers motivate careful thought and truthful reporting, even for unverifiable survey items?

We develop and test two simple, psychologically grounded methods to improve response quality for both verifiable and unverifiable questions. Our first approach, the *Bonus* method, provides respondents with an unexpected payment midway through the survey. Drawing on evidence that reciprocity powerfully motivates behavior (Fehr and Gachter, 2000; Fehr et al., 2002), we hypothesize that the bonus can induce reciprocal effort. Although gift-exchange effects are well documented in labor markets (Akerlof, 1982; Gneezy and List, 2006; Falk,

[1] See Schotter and Trevino (2014), Schlag et al. (2015), and Barseghyan et al. (2018) for comprehensive review articles on eliciting beliefs and risk preferences.

2007), their usefulness for improving survey response quality is largely unknown.[2]

In our second approach, the *Restatement* method, we tell participants that they will have to restate their (previously-given) answers to a random subset of questions later in the survey and will be paid for correct restatements. This relies on a key insight that "truths" are fundamentally easier to reproduce. For stable preferences, long-term memories, and reproducible cognitive processes, careful initial answering facilitates accurate restatement *without* requiring memorization. For instance, respondents need not memorize their favorite book or the answer to 21+35 to repeat these answers later, but arbitrary answers may be harder to reproduce. Anticipating this, respondents may exert greater initial effort.

The goal of this paper is not to immediately promote the use of these methods. Both methods raise some obvious concerns. For example, the Bonus method could fail to induce reciprocity, e.g., if it crowds out intrinsic motivation or if the bonus does not come as a surprise due to its widespread adoption (DellaVigna et al., 2022). The Restatement method could backfire if participants resort to memorizing answers or if they "game the system" by adopting low-effort strategies, e.g., choosing salient but incorrect answers or copying answers down. In addition, there is no reason to believe that these methods should reduce social desirability bias or experimenter demand, which are other crucial challenges when conducting surveys.[3] The primary aim of the paper is to investigate whether these interventions rooted in psychological factors—specifically reciprocity and the cognitive ease of restating truths—have any potential to improve survey quality as a first step. Our hope is that the resulting insights lead to further tests and improved methods.

We test the methods using a large online experiment (N=2,428) on Prolific. Midway through the survey, participants are randomly assigned to one of five conditions. The first

[2] A large literature tests the effect of varying monetary payments on mail survey *response rates*, but typically does not test bonus payments introduced in the middle of a survey (Yammarino et al., 1991).

[3] We discuss these concerns more in Section 2. In this experiment, we avoid question types prone to these biases to focus on the primary goal.

two involve our novel interventions: (i) *Bonus*, where participants receive an unexpected payment framed as a gift; and (ii) *Restatement*, where participants are informed they can earn payments for accurate restatements. We benchmark these against: (iii) *Conventional Incentives* offering payments for objectively correct answers; and (iv) *No Incentives*, a control condition. We also evaluate a variant called (v) *Restatement-Points*, where restatement rewards are non-monetary points. This is to examine whether the method could work with only intrinsic incentives. All participants were recruited at the same posted base wage, and payments were calibrated to equalize expected total earnings across groups.

We have two main measures of answer quality. First, we assess *correctness* for verifiable questions with objectively correct answers, including Raven's matrices, simple calculations, and choices between lotteries with a stochastically dominant option. Second, to address the fundamental challenge of measuring response quality for unverifiable responses, we measure internal *consistency*, capturing whether respondents give compatible answers to pairs of differently-worded versions (near-duplicates) of the same question.[4] This follows a literature on test-retest reliability in clinical measurement (Bland and Altman, 1986), but uses it to assess respondent reliability rather than question reliability.

The results show that the Bonus method yields modest improvements in response quality. Consistency increases by 3 p.p. (5%) and correctness improves by 2 p.p. (3%) in our preferred specification. These gains arise despite total payments matching those paid to the control group. Compared to dropping participants who fail an attention check, the bonus has a larger impact on improving consistency (p=0.10), and a similar impact on correctness (p=0.29). Measures of behavior also indicate increased effort: longer response times, greater restatement accuracy (despite no anticipation of restatement), and higher self-reported attention. The effects are concentrated on easier questions, where additional effort

[4]For example, we ask for the proportion of U.S. income going to the top 20% of earners in one question and the bottom 80% in another. Consistent answers sum to 100%.

is more likely to enhance accuracy.[5]

The Restatement method substantially increases internal consistency by 8.4 p.p. (14%). However, the method has little effect on correctness—if anything, there is a small negative impact despite increases in response time and self-reported effort. The contrast between its large positive impact on consistency and its null effect on correctness presents a puzzle: Does the method genuinely enhance answer quality? Or does it merely encourage participants to memorize their answers and therefore better recognize near-duplicate questions? In the latter case, participants may simply be reproducing their prior answers,[6] in which case consistency does not reflect higher answer quality.

To distinguish between these two explanations, our supplementary experiment (N=1,408) tests whether the Restatement method improves initial response quality in the absence of near-duplicate questions. We replace our consistency questions with pairs of questions measuring *correlations* between closely related beliefs and preferences. Each pair comes from the Pew American Trends Panel Survey (Pew Research Center, 2024), with one question concerning a participant's belief and the other a policy preference about a particular topic.[7] Answers to questions in a pair are highly correlated in both the original Pew data and our data. More thoughtful responses should strengthen these natural correlations by reducing measurement error. However, we find that the Restatement method does not enhance correlations. We therefore conclude that the method does not improve answer quality for the average participant.

To unpack the impact of the Restatement method, we consider a multitasking principal-agent framework in which respondents can increase restatement success by exerting effort

[5]These modest gains also suggest that accuracy may be relatively insensitive to effort. Even the *Conventional Incentives* treatment, which pays directly for correct answers, does not significantly increase correctness, consistent with results from Danz et al. (2022); Enke et al. (2023); Mason and Watts (2009).

[6]Note that there is *no direct incentive* to be consistent in the *Restatement* treatment or any treatment. However, participants may try to be consistent as a means to increase the restatement payment.

[7]For example, we measure the correlation between (i) belief in the threat of climate change and (ii) preferences for climate change-mitigating policies.

along two dimensions: (i) answering carefully, which improves initial answer quality, or (ii) memorizing answers or "gaming," which does not affect or harms initial answer quality. A debrief question in the main experiment asking about participants' reactions to the *Restatement* method demonstrates that both channels are important. While few report writing down answers or choosing salient incorrect answers, 25.0% of participants report trying to memorize answers.[8] Among these, the method leads to significantly lower correctness. However, a much larger share (52.6%) report trying to answer carefully and show improved consistency as well as correctness. This heterogeneity explains the overall null effect on correctness, which we view as an important scientific finding regarding the null (Abadie, 2020).

Our findings demonstrate that framing mid-survey payments as unexpected gifts can modestly improve data quality, even for unverifiable questions and particularly for easier questions. Our study also reveals important challenges in using a method that relies on participants recognizing that it is easier to reconstruct the truth: incentives can lead to multitasking concerns, whereby some participants focus on other dimensions of effort such as memorizing answers. The method is nevertheless consistent with improvements in response quality among a subset of participants, suggesting that refined versions of this approach could generate overall improvements.

Related Literature. Our primary contribution is to expand the nascent toolkit for incentivizing high-quality responses to unverifiable survey questions. One strand of prior work, which includes the Bayesian Truth Serum and its variants (Prelec, 2004; Witkowski and Parkes, 2012; Radanovic and Faltings, 2013; Baillon, 2017), peer prediction (Miller et al., 2005), Bayesian markets (Baillon, 2017), choice matching (Cvitanic et al., 2019), and peer betting (Baillon et al., 2025), models respondents as players in a game and establishes truth-telling as a Bayesian Nash equilibrium. Such approaches can be effective (e.g. Baillon

[8]Explicit gaming reportedly affects less than 5% of questions, and dropping all such participants does not affect the main results. As discussed in Sections 2 and 3, we avoided including questions with salient answer options or Likert scales, which we suspect mitigated the use of gaming strategies.

et al., 2022), but they all require that agents are strategically sophisticated and Bayesian for incentive compatibility, assumptions that may not hold in practice.[9] By contrast, we introduce two simpler mechanisms, rooted in psychological principles, that do not require any strategic or Bayesian reasoning. The instructions are simple to convey in natural language and only negligibly increase the length of the survey—both important considerations for "behavioral incentive compatibility."[10]

A few recent papers are more in line with our psychological approach. Jacquemet et al. (2019) show that people perform better in discrete choice tasks if they first sign an oath to "faithfully and conscientiously fulfill the tasks."; and Corgnet and Hernán González (2023) find that, due to regret motives, randomizing piece-rates or performance targets motivates performance better than deterministic schemes.

We also contribute to the broader literature on improving survey data quality (Stantcheva, 2023) by developing techniques that directly enhance participant attentiveness rather than filtering out inattentive responses. Current best practices rely heavily on attention checks (Huang et al., 2012; Berinsky et al., 2014; Curran, 2016; Berinsky et al., 2016; Abbey and Meloy, 2017), but these methods risk annoying respondents, increasing attrition, and creating selection bias when used to exclude respondents. Despite not requiring any additional payments, we find that our Bonus method improves answer quality more than dropping participants who fail an attention check. Other "ex-post" corrections involve collecting multiple measures of the same quantity. For instance, this can be used to reduce classical measurement error (e.g. Gillen et al. (2019) and Luttmer and Samwick (2018)) or optimally weight participant responses to improve the explanatory power of regression models (Falk et al., 2021). However, these methods may substantially increase survey length.

[9] Benjamin (2019) reviews the literature on failures of Bayesian updating.

[10] Danz et al. (2022) show that when the details of the binarized scoring rule (Hossain and Okui, 2013) are made salient, participants give *less* accurate answers on very simple questions, suggesting that participants may be confused by seemingly artificial incentive schemes.

Finally, our paper builds on important work in psychology and behavioral science. Our Bonus method is based on the well-known "gift exchange" effect (e.g. Fehr and Gachter, 2000), whereby subjects reciprocate a "gift" by putting in more effort. The Restatement method builds on a literature that explores the relationship between truth-telling and memory, particularly work on "judgements of learning" that researches people's own predictions of their ability to recall (Benjamin et al., 1998; Vieira and Lane, 2012; Besken, 2018; Zimmermann, 2020). Our novelty is to apply these insights to improve data quality in surveys.

2 Conceptualizing new methods

We conceptualize two novel methods for increasing survey response quality (i.e., accuracy). Before starting a key survey section (not necessarily the first), subjects are informed that the methods will be applied to the questions in that section.

In the *Bonus* treatment, subjects learn about an unexpected monetary bonus $B > 0$, framed as a bonus to encourage attention and effort. We hypothesize that this increases the intrinsic motivation to answer accurately due to a desire to reciprocate (as per the evidence on reciprocity in psychology, e.g., Fehr and Gachter, 2000).

In the *Restatement* treatment, subjects learn that they will observe a random subset of questions again later in the survey (i.e., repeat questions Q_R), and will be paid $R > 0$ for correctly restating their previously-given answers. This exploits a simple logic: careful initial responses may be cognitively easier to reconstruct later, while careless responses may require significant memorization effort in order to restate. Thus, for questions with reconstructible answers—e.g., concerning stable underlying preferences, long-term memories, or calculations—answering truthfully can be a more efficient route to correctly restating an answer.

In Appendix B, we illustrate how the *Bonus* and *Restatement* treatments may affect

behavior using a common theoretical framework. To provide a quick summary, participants derive both intrinsic utility from answering accurately and extrinsic utility from total payments. Exerting effort toward answering carefully increases accuracy, but is cognitively costly. If the *Bonus* treatment increases the intrinsic payoff for accuracy, then participants will increase effort and therefore accuracy.[11] In the *Restatement* treatment, we posit that the probability of restatement success on repeat questions (Q_R) depends on two dimensions of effort: participants may exert effort toward *providing careful initial answers* and/or *memorizing their answers or gaming*. Both increase the probability of restatement success, but only the former increases initial answer accuracy. The restatement payment R will therefore increase the former type of effort if (i) providing careful initial answers makes it easier to reconstruct answers and (ii) providing careful initial answers is relatively "cheap" compared to memorizing or gaming. However, if these conditions are violated, the Restatement method can backfire.

Limitations and scope. We do not presume that the Restatement or Bonus methods reduce social desirability bias or experimenter demand effects (Bertrand and Mullainathan, 2001; De Quidt et al., 2018). Indeed, for questions that evoke such concerns, these effects may be reinforced, leading to a reduction in accuracy. As our framework highlights, the Restatement method may also backfire whenever there is any particularly salient answer, which could motivate the gaming strategy of choosing this answer and (easily) committing it to memory. As our experiment is the first test of whether these methods can work, we avoid including questions that may evoke these types of concerns. In particular, we avoid Likert-scale questions where subjects may be tempted to choose first or last answers, as well as numeric answer options with salient numbers such as 0. To further mitigate the attractiveness of gaming, we disable text-copying and inform subjects in the *Restatement*

[11] A *conventional incentive* in the form of providing a payment for accuracy increases extrinsic returns to effort, but this is only possible with verifiable questions.

treatments that the order of answer options in repeat questions will be shuffled.

3 Experimental design

Recruitment. We recruited 2,428 US-based adults on Prolific, an online survey platform, between January and February 2023. Prolific is known to have high data quality relative to similar platforms (Peer et al., 2022), suggesting that our effects are likely to be lower bounds relative to those from other platforms. The survey took 38.6 minutes to complete on average. All participants were recruited with the same advertised completion fee of £4, where £1 ≈ $1.25. The full questionnaires are available here and here.

3.1 Treatments

The experiment consisted of five between-subject treatments, administered at the start of the main section of the survey.

1. **Bonus** (N=613). Participants were told they would receive an "extra bonus" payment of £3 for completing the survey.

2. **Restatement** (N=599). Participants were told about the repeat questions, i.e., that they would receive, for each of three randomly selected questions from the main section, £1 for correctly restating their previous answer (up to £3 total).

3. **Restatement-Points** (N=300). Similar to the above, but participants received 1 "point" (with no monetary value) per correctly restated answer.

4. **Conventional Incentives** (N=307). Participants were told they would be paid based on 3 randomly selected questions from the main section. If the question had an objectively correct answer, they would receive £1 for giving the correct answer. If the

question asked for a preference (e.g., over lotteries), they would receive their chosen alternative.

5. **No Incentives (Control)** (N=609). No additional information was given.

The language describing the incentive schemes was kept as similar as possible across treatments. To reduce differential experimenter demand, all instructions (including control) included the phrase: "we would like you to pay attention and put effort into answering the questions in this section." Comprehension checks confirmed that participants understood their incentive scheme. Completion fees were adjusted so that expected total payments were approximately equal across arms (£6.70–7.27), providing a conservative test in which only the timing and framing of payments differs.[12]

3.2 Survey sections

Figure A.1 summarizes the overall experiment structure, which involved the five following stages:[13]

1. **Background.** This stage contained 30 unincentivized demographics questions. As our methods target the inattention and carelessness that increase with survey fatigue, this lengthy initial stage was designed to tire participants before treatment, while also masking the study's purpose. This stage included an attention check, but we did not drop those who failed it.

2. **Main.** At the start of this stage, participants received the instructions relevant to their assigned treatment. They then answered 14 *correctness* questions and 8 *consistency*

[12]Those in the *No Incentives* and *Restatement-Points* treatments received a higher completion fee of £7 (vs. £4 for other treatments). All participants clicked a button to acknowledge the completion fee, but it was not emphasized that it was higher than advertised in these treatments. Mean total payments were: £7.27 for *Bonus*, £7.26 for *No Incentives*, £7.11 for *Conventional Incentives*, £6.70 for *Restatement*, and £7 for *Restatement-Points*.

[13]The full questionnaires are available here and here.

questions (described in Section 3.3), largely taken from existing surveys to ensure relevance to practitioners. Questions were randomly ordered at the participant level.

3. **Spillover.** To test whether the effects of incentives in the main stage persist (or "spill over") once the incentives are removed, we included 10 additional questions that were explicitly unincentivized. These included 3 correctness questions, 1 pair of consistency questions, and an attention check.

4. **Repeat Questions.** In all treatments, 3 randomly selected questions from the main stage were shown and participants asked to restate their previously-given answers. *Restatement* and *Restatement-Points* participants were already informed about this before the Main stage; for other treatments it came as a surprise.[14]

5. **Debrief.** The final stage included questions designed to understand the mechanisms behind any treatment effects, including self-reported reciprocity toward the researchers, changes in attention and effort, and (for *Restatement* participants) the strategies used to prepare for the repeat questions.

3.3 Outcomes and specification

Our primary outcomes are two measures of answer quality—correctness and consistency.

Correctness. Correctness is an indicator for whether respondent i gave the objectively correct answer to question j. The correctness questions (Table A.1) are designed to be plausibly responsive to effort, rather than pure knowledge questions. They include simple calculations, Raven's Matrices, items from the cognitive reflection test (Frederick, 2005),

[14] *Restatement* participants received £1 per correctly restated answer, as announced at the start of the Main stage. In the *Bonus* and *Conventional Incentives* treatments, participants received £0.10 per correctly restated answer to keep total payments similar across arms. *No Incentives* participants were randomized to receive either £0.10 or £1 per correct restatement: the £0.10 group maintains payment parity, while the £1 group allows us to compare restatement performance when the repeat stage is anticipated (*Restatement*) versus unanticipated (*No Incentives*), holding fixed the restatement payment.

dot-counting (Dewan and Neligh, 2020), reading comprehension exercises, and choices from lotteries with first-order stochastically dominant options.[15]

Consistency. Consistency is an indicator of whether respondent i gave compatible answers across a pair of near-duplicate questions j. Unlike correctness, this measure can be applied to unverifiable questions with no objectively correct answer. The 4 consistency pairs (Table A.2) are reworded or obfuscated near-duplicates concerning beliefs about inequality, education, and immigration, mimicking common survey questions (Alesina et al., 2018). For example, participants estimate the income share of the top 20% of earners (question 3a) and the bottom 80% (question 3b); consistent answers sum to 100%. If preferences and beliefs are stable during the survey, variation across elicitations reflects measurement error, so greater consistency implies higher answer quality.[16]

Verifiability of consistency questions. To test whether our methods have different effects on verifiable and unverifiable questions, we randomly assigned participants to answer either verifiable or unverifiable versions of the consistency questions.[17]

Empirical specification. Following our pre-registration, we analyze correctness and consistency separately, pooling all questions in each category into an OLS regression:

$$Y_{ij} = \beta_1 Bonus_i + \beta_2 Restate_i + \beta_3 RestatePoints_i + \beta_4 Conventional_i + \mathbf{X}_i'\Gamma + \mu_j + \varepsilon_{ij} \quad (1)$$

where i is the participant, and j is the question (for correctness questions) or question-pair (for consistency questions). $Y_{ij} = 1$ if the respondent answers correctly or consistently, and

[15] We also included 2 exploratory Bayesian updating questions, preregistered to be excluded from the main analysis given evidence that even strong incentives cannot improve Bayesian updating (Enke et al., 2023). See Table A.1 notes.

[16] This follows work on test-retest reliability in clinical measurement (Bland and Altman, 1986) and the practice of using multiple measurements to reduce measurement error (Gillen et al., 2019).

[17] For instance, some estimated the income share of the top 20% of U.S. earners **in 2050** (unverifiable), while others estimated it **in 2023** (verifiable). Only verifiable questions were used in (i) *Conventional Incentives* so that we could always pay participants for objectively correct answers and (ii) *Restatement-Points* to reduce costs.

$Y_{ij} = 0$ otherwise. *No Incentives* is the omitted category. μ_j are question (or pair) fixed effects, and standard errors are clustered at the participant level. In some specifications, we include a vector of controls $\mathbf{X}_i$ selected by double LASSO (Belloni et al., 2014) to account for confounding imbalances (see Appendix D.2 for list of controls selected).

Additional outcomes. To understand mechanisms, we also examine response times, self-reported effort and attention, reciprocity toward the researchers, and performance in the Repeat Questions stage. Each is described when we present the results.

Pre-registration. The study was preregistered on AsPredicted (main experiment and supplementary experiment). The only deviations are the inclusion of LASSO controls in some specifications, justified below, and pooling verifiable and unverifiable questions in the main specification (since unpooled results are very similar; see Table A.7).

3.4 Sample characteristics, balance, and attrition

Table A.3 shows participants' average demographic characteristics in all treatments. We find no significant overall imbalances between the *No Incentives* group and the *Conventional Incentives*, *Restatement*, or *Restatement-Points* groups (p of joint F-tests: 0.32, 0.39, 0.87). However, there is a random but meaningful imbalance in the *Bonus* group (p of F-test = 0.05), which generates a downward bias on our estimates: predicted correctness and consistency scores based purely on demographics are significantly lower in this arm (Table A.4, columns 3–8). This imbalance is not driven by differential attrition, which is small overall (3.2% from the start, 1.4% after demographics) and does not differ across treatments (Table A.4, columns 1–2). To account for this imbalance, our preferred specifications include controls selected by double LASSO.

4 Main results: impact on correctness and consistency

Table 1 and Figure 1 show the effects of the treatments on correctness and consistency.

The *Bonus* treatment improves consistency and, in our preferred specification with LASSO controls, correctness. Consistency improves by approximately 3 p.p. (Table 1, column 4, 5%, p=0.02). Correctness improves by a more modest 2.1 p.p. (column 2, 2.8%, p=0.03) in the specification with LASSO controls.[18]

Restatement and *Restatement-Points* both have large positive impacts only on consistency. Participants are 8.4 p.p. more likely to give consistent answers in the *Restatement* arm (columns 3-4, p<0.001), and 5 to 7 p.p. more likely in the *Restatement-Points* arm (p<0.001). By contrast, neither has a detectable effect on correctness (columns 1-2, p=0.19 and 0.15 respectively), and the coefficient estimates are slightly negative. We are well-powered to rule out even relatively small effects, with an implied minimum detectable effect (α=5%, β=80%) of ±3.1 p.p. (4.1%) in the preferred LASSO specification. This is thus clear evidence that correctness does not improve.

These results contrast with those of *Conventional Incentives*, which has no detectable effect on either consistency or correctness. Despite their common use, such incentives have limited impact on survey response quality, echoing recent findings in the literature (e.g. Enke et al. (2023)).

Robustness. The results are robust to (i) adjusting for multiple hypothesis testing (Table A.5, columns 1 and 5); (ii) using randomization inference (columns 2 and 6); (iii) using an inverse-covariance weighted index (Anderson, 2008) (columns 3 and 7); and (iv) dropping the 11.1% participants who failed an attention check (columns 4 and 8). The improvements persist among those who passed the attention check, suggesting that outcomes

[18] Without these controls, the effect is no longer significant (Table 1, column 1, p=0.59), but this reflects sample composition rather than treatment effects. The *Bonus* group started with demographic characteristics that predict 1.4 percentage points fewer correct answers (see Table A.4, columns 3-5). We prefer the LASSO specification because it better accounts for this baseline imbalance.

improve across the distribution of respondents, not only among the most inattentive.[19]

Heterogeneity by question difficulty. Figure A.2 presents effects for each question individually. The *Bonus* effect on correctness is concentrated on easier questions: for questions where control-group correctness exceeds 74.8% (the median), *Bonus* increases correctness by 3.7 p.p. (Table A.6, column 1, p=0.01), with no effect on harder questions. This is relevant for practitioners: the bonus is most helpful for questions where additional effort can plausibly improve accuracy.[20]

Heterogeneity by verifiability. Treatment effects do not vary by whether consistency questions are verifiable or unverifiable (Table A.7), suggesting that the methods' efficacy does not depend on verifiability.[21]

Spillover. A natural concern is that the incentives could exhaust or frustrate participants, reducing answer quality once the incentives are removed. We find the opposite: effects persist in the unincentivized Spillover stage (Table 1, columns 5–6). The *Bonus* improves correctness by 2.9 p.p. (p=0.07) and *Restatement* improves consistency by 7.8 p.p. (p<0.001), matching the magnitudes in the Main stage.

Comparison to attention checks. The *Bonus* treatment compares favorably to the standard practice of dropping participants who fail an attention check. Dropping the 11.1% who fail increases correctness by only 1.1 p.p. and consistency by 0.7 p.p. (Table A.8). *Bonus* produces 4.2 times larger gains in consistency (p of difference: 0.10) and similar improvements in correctness (p=0.29), without incurring additional costs or discarding any data.[22]

[19] We also find no heterogeneity by baseline response time in the Background stage (Table A.6, columns 2–5).

[20] The *Restatement* treatment also produces heterogeneous effects across correctness questions (Figure A.2, p=0.03), but this is not systematically explained by question difficulty (Table A.6, column 1).

[21] The consistency effects are also unaffected by whether the near-duplicate pair appears early or late in the survey or if the questions in a pair appear closer together (Table A.7, columns 3–4).

[22] Those who pass the attention check are 6.4 p.p. (p=p<0.001) more correct and 4.8 p.p. (p=0.001) more consistent than those who fail, in line with other work (Alvarez and Li, 2021).

4.1 Supplementary experiment: does Restatement genuinely improve answer quality?

The *Restatement* treatments have a large impact on consistency but no impact on correctness. Does this reflect a genuine improvement in answer quality? Or does the method simply induce participants to memorize their answers, better recognize near-duplicates, and choose to act in a consistent manner without actually improving answer quality?[23]

To distinguish these interpretations, we ran a supplementary experiment (N=1,408) that replaces near-duplicate consistency pairs with pairs of *correlated* questions. Each pair contains one question eliciting a belief and another eliciting a policy preference on the same topic (e.g., perceived threat of climate change paired with support for climate action). These questions are distinct, so participants cannot simply repeat a memorized answer, but they are naturally correlated in both national survey data and our sample. If the Restatement method genuinely improves answer quality, it should strengthen these correlations by reducing measurement error. We only consider the comparison between *Restatement-Points* and *No Incentives* as the former increased consistency in our original experiments, and this maintains identical monetary payoffs across treatments. Details are in Appendix C.

Comparing *Restatement-Points* to *No Incentives*, we find no significant impact on the correlation measure, despite the large sample.[24] This suggests that the consistency gains in the main experiment do not reflect improved answer quality. Instead, the Restatement method appears to compel some participants to exert effort on memorization rather than answering carefully, a mechanism we explore in Section 5.3.

[23]Note that there is *no direct incentive* to be consistent in the *Restatement* treatment or any treatment. However, participants may try to be consistent as a means to increase the restatement payment or have a preference for consistency per se.

[24]We also replicate the null on correctness without potentially distracting near-duplicate questions (p=0.48, Table A.10, columns 1–2).

5 Mechanisms

We now explore the mechanisms behind the *Bonus* and *Restatement* results, documenting impacts on effort, reciprocity, and memorization.

5.1 Other measures of effort

Response time (Figure 2, panel A). The *Bonus* treatment increases average response time per question by 3.4 seconds (10%, $p<0.01$), while the *Restatement* treatment increases it by 2.2 seconds (6%, $p=0.03$), in our preferred specification controlling for baseline response times.[25] The effects are similar across consistency and correctness question types, suggesting that both treatments lead participants to exert more effort overall.

Repeat Question performance (Figure 2, panel B). The *Restatement*, *Restatement-Points*, and *Bonus* treatments all increase the fraction of correctly restated answers from 87% to 90–91% (2.8–3.9 p.p.). That the *Restatement* treatments improve restatement performance is consistent with the proposed mechanism: participants exert more effort in the main stage, which facilitates later restatement. The *Bonus* result is also revealing: because the Repeat Questions stage came as a surprise for these participants, their improved performance likely reflects greater initial effort when answering questions.[26]

Self-reported effort (Figure 2, panel C). In the Debrief stage, we asked participants how their incentive scheme affected their *attention* and *effort*, and whether they answered as quickly or as carefully as possible on a 0-5 scale. All treatments led to increases in a combined effort index of 0.39–0.60 SD (all $p<0.001$) relative to *No Incentives* where participants were first reminded that they received a higher-than-advertised completion fee (£7 vs. £4).

[25] The *Restatement* effect on response time is no longer significant without the baseline control (1.6 seconds, $p=0.12$). *Conventional Incentives* also increase response time, but *Restatement-Points* does not have a significant effect. All results are robust to Winsorizing outliers.

[26] In the *No Incentives* group, offering £1 versus £0.10 for correct restatements does not affect performance ($p=0.13$), suggesting that initial effort matters more than the current restatement payoff.

5.2 Bonus: Reciprocity

We construct a reciprocity index from two debrief measures: self-reported gratitude toward the researchers and willingness to answer 3 optional questions as a favor (Figure 2, panel D). Only the *Bonus* treatment significantly increases reciprocity, primarily through increased gratitude. The *Restatement* treatments, by contrast, reduce reciprocity, possibly because participants are frustrated by longer survey times or by payments conditional on their behavior (Stantcheva, 2023).

5.3 Restatement: answering carefully vs. memorizing or gaming

The results above suggest increased effort under the *Restatement* treatments, despite the effect being limited to increased consistency on near-duplicate questions. As our conceptual framework (Section 2) highlights, this effort could either improve or harm answer quality, depending on whether participants direct it toward answering carefully on the one hand, or memorizing or gaming on the other. We use debrief questions to examine these channels.

Gaming. Gaming behavior is rare and does not explain our findings. Among *Restatement* participants, only 14.5% report choosing easy-to-remember answers, 5.8% report writing answers down, and 4.3% report searching online. Moreover, participants who reported these strategies claimed doing so only for a low proportion of questions (suggesting less than 5% of questions were affected overall). Excluding these participants does not alter the main results (Table A.11).[27]

Answering carefully vs. memorizing. In an open-ended debrief question, we asked *Restatement* participants to describe their strategies for the Repeat Questions stage, and coded responses using GPT-4o-mini (see Appendix D.1). For example, one participant wrote,

[27] In the supplementary experiment (October–December 2024), only 1.8% of respondents reported using ChatGPT. *Restatement* participants also do not select first or last options more often in randomly ordered response lists.

"I read the questions slowly and carefully several times before answering" (coded as answering carefully); another wrote, "I tried hard to remember my answers so I could restate them later" (coded as memorizing). Overall, 52.6% of participants reported answering more carefully and 25.0% reported trying to memorize their answers.[28]

Heterogeneity by strategy. Performance varies substantially with participants' reported strategies (Figure 3). Participants who report answering more carefully achieve both higher correctness (β=4.3 p.p., p<0.001) and consistency (β=10 p.p., p<0.001) relative to *No Incentives*. Memorizers, by contrast, perform 3.6 p.p. worse on correctness (p=0.04), suggesting that memorization crowds out careful answering. Memorizers also show no overall improvement in consistency, though they do achieve slightly higher consistency on simple reworded duplicates (β=3.5 p.p., p=0.06, questions 2 and 4 in Table A.2) but not on questions requiring attention to reversed wording (p=0.60, questions 1 and 3), suggesting superficial recall rather than careful attention.

These results are correlational and should be interpreted with caution, but the pattern is consistent with improved answer quality for some participants and a reduction for others, producing the overall null. This aligns with the prediction from our conceptual framework that Restatement incentives can have contrasting impacts depending on how participants direct their effort.

6 Conclusion

This paper tests two psychological mechanisms for improving survey response quality. Both methods apply to unverifiable questions, are simple to explain in natural language, and add minimally to survey length. The *Bonus* method improves both correctness and consistency by 2–3 percentage points, driven by reciprocity toward the researchers, and compares fa-

[28]Only 0.7% of respondents were classified as gaming; 32.7% as using no particular strategy; and 0.3% as using some other strategy.

vorably to the standard practice of dropping participants who fail attention-checks. The *Restatement* method does not improve average answer quality despite increasing consistency by 8 percentage points: it falls prey to a multitasking problem in which some participants reallocate effort toward memorizing answers instead of answering carefully. On the other hand, conventional incentives have no detectable effect, underscoring the difficulty of improving data quality through standard approaches.

Our findings demonstrate both the promise and challenges of psychological incentives in surveys. The success of the Bonus method relative to conventional incentives suggests the possibility of extending it in several directions. First, future work could explore the practicalities of implementing the Bonus method. Can it work for in-person surveys or repeated waves? How does it interact with other methods, and what is the optimal timing within a survey? Second, researchers should investigate how to amplify the bonus effect, e.g., by combining it with norms messaging such as accuracy pledges.

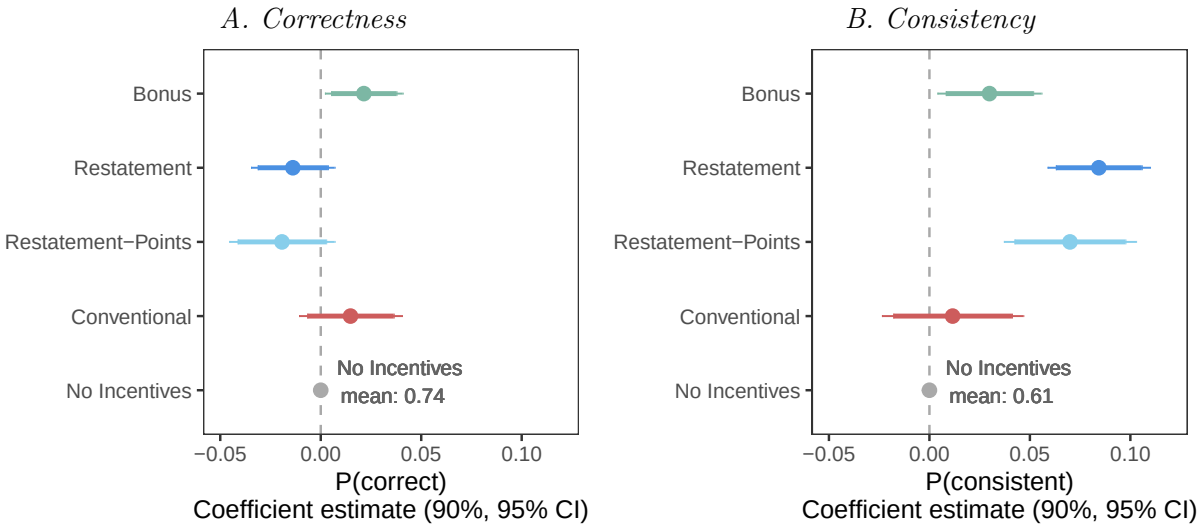
The fact that around half of the participants report responding to the Restatement method by answering more carefully—with corresponding increases in consistency and correctness—is also promising. Future research could explore refined methods that minimize the memorization motive, e.g., informing participants that they will later have to *explain their reasoning process* behind a previous answer, and evaluating the coherence between the explanation and initial response. Ultimately, our results show that psychology-based methods can effectively increase effort; however, more work is needed to better direct this effort toward the goal of improving response quality.

Table 1: *Impact on correctness and consistency*

	Main stage				Spillover stage	
	Correct Answer (=1)		Consistent Answer (=1)		Correct Answer (=1)	Consistent Answer (=1)
	(1)	(2)	(3)	(4)	(5)	(6)
Bonus	0.006	0.021**	0.026*	0.030**	0.029*	0.009
	(0.012)	(0.010)	(0.014)	(0.013)	(0.016)	(0.019)
Restatement	−0.020	−0.014	0.086***	0.084***	0.014	0.078***
	(0.013)	(0.011)	(0.014)	(0.013)	(0.016)	(0.019)
Restatement-Points	−0.020	−0.019	0.054***	0.070***	0.002	0.121***
	(0.016)	(0.013)	(0.017)	(0.017)	(0.021)	(0.025)
Conventional	0.011	0.015	0.000	0.012	0.029	0.031
	(0.014)	(0.013)	(0.018)	(0.018)	(0.021)	(0.025)
Num. observations	33 992	33 992	9712	9712	7284	2428
Num. participants	2428	2428	2428	2428	2428	2428
Dep. var. control mean	0.74	0.74	0.61	0.61	0.71	0.83
R-squared	0.07	0.14	0.22	0.25	0.10	0.06
Question FEs	X	X	X	X	X	
LASSO controls		X		X	X	X
p-val: Bonus=Conv.	0.74	0.61	0.14	0.31	1.00	0.38
p-val: Restate.=Bonus	0.03**	0.00***	0.00***	0.00***	0.35	0.00***
p-val: Restate.=Conv.	0.04**	0.03**	0.00***	0.00***	0.47	0.05*
p-val: Restate.=Restate-Points	0.99	0.69	0.06*	0.38	0.57	0.09*
p-val: Restate. (Mon. + Points pooled)=0	0.07*	0.11	0.00***	0.00***	0.48	0.00***

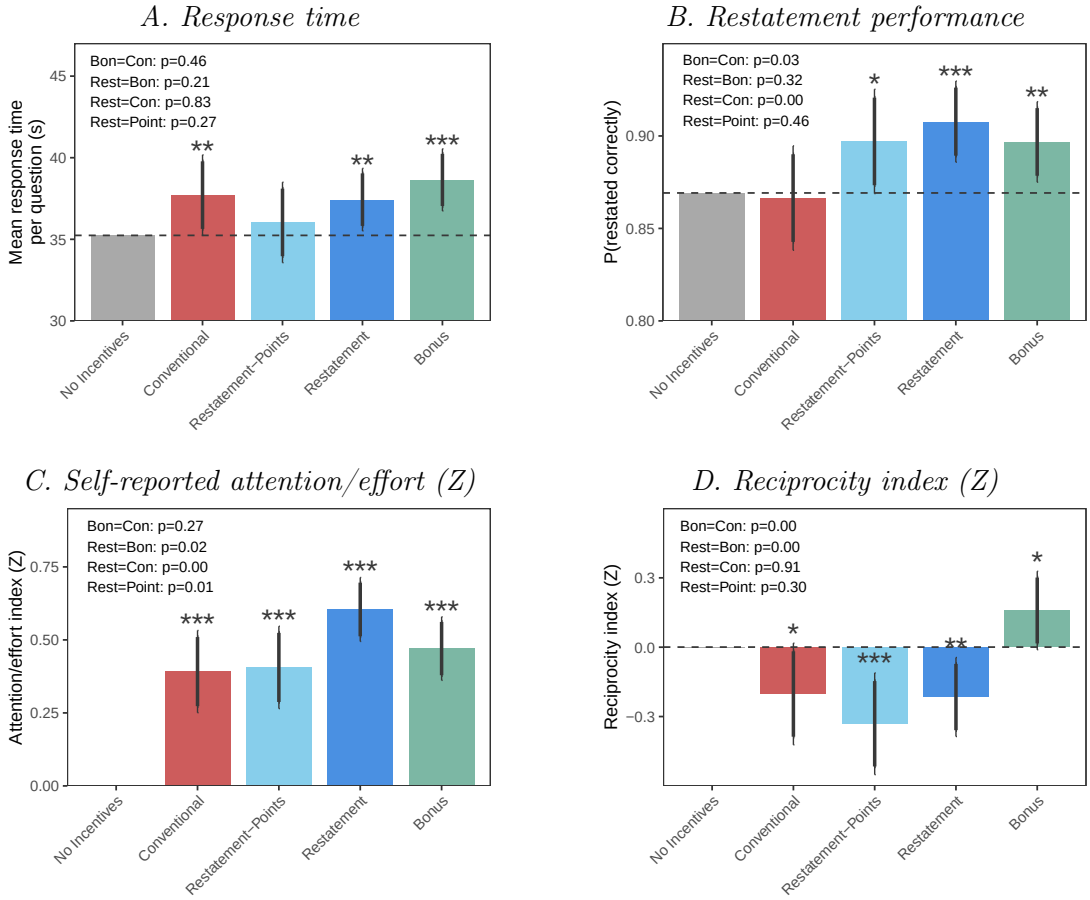
Notes: Uses specification from Equation 1. Observation is the (participant × question) for columns 1-2 and 5, (participant × question-pair) for columns 3-4, and participant-level for column 6. Omitted category is the *No Incentives* arm. Outcome is a binary indicator for a correct answer in columns 1-2 and 5, and a binary indicator for a consistent pair in columns 3-4 and 6. Columns 1-4 use the Main stage; columns 5-6 use the unincentivized Spillover stage. Selected LASSO controls are in Appendix D.2. Standard errors clustered at the participant level. Only verifiable consistency questions were used for *Conventional Incentives* (so that participants could always be paid for correct answers) and *Restatement-Points* (to reduce survey costs); the sample size for these treatments is therefore approximately 300 instead of 600.

Figure 1: *Plotting the estimated treatment effects*



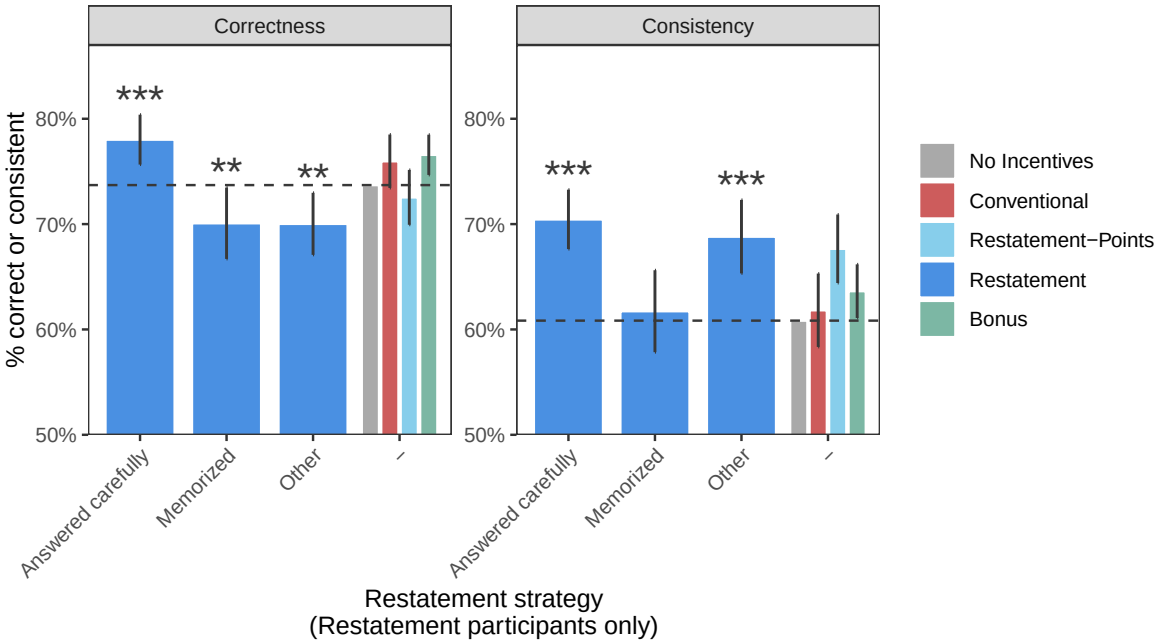
Notes: Panel A is from column 2 of Table 1; panel B is from column 4. Both use LASSO controls and question fixed effects (see Appendix D.2).

Figure 2: *Impact on effort and reciprocity*



Notes: Panel A (*Response time*): outcome is the mean response time for questions in the main stage. Panel B (*Restatement performance*): outcome is the probability of correctly restating for the 3 restatement questions. Panel C (*Self-reported attention/effort*): participants were asked how their incentive scheme affected how much *attention* they paid (from *Greatly decreased* (1) to *Greatly increased* (5)), and a similar question for how much *effort* they exerted; and whether they answered as quickly as possible or as carefully as possible on a 0-5 scale. We normalize each question to have mean of 0 and standard deviation of 1 in the *No Incentives* (control) group, then the outcome is the normalized average of all 3 questions. Panel D (*Reciprocity index*): participants were asked how thankful they were for their payment (scale 1-5) and whether they were willing to answer three more questions at the end of the survey (yes/no). Outcome is the average of the normalized Z-score of both questions, with mean of 0 and standard deviation of 1 in the control group. All models include all controls selected by double LASSO for both consistency and correctness measures. Thick bars are 90% confidence intervals, thin bars are 95%, calculated relative to control group (*No Incentives*). Stars indicate differences with *No Incentives* (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$). Panel A (*Response time*) is an ANCOVA specification that controls for baseline response time in the demographic stage.

Figure 3: *Heterogeneity by strategy used in Restatement arm*



Notes: 95% confidence intervals relative to *No Incentives* (control) group. Restatement strategy was only asked to *Restatement* participants using an open-ended text-response question: "What was your main strategy for making sure you could correctly restate your answers?" We used OpenAI's API to code participants' responses, noting whether they answered carefully, memorized, or followed some other strategy (participants could be coded as following multiple strategies, see Appendix D.1). In the *Restatement* arm, 315 participants answered carefully, and 150 memorized (57 did both). 210 participants are classified as "Other", of which 196 had no particular strategy. Participants can be in multiple columns, since they can follow multiple strategies. Outcomes and specifications are otherwise the same as in Table 1.

References

Abadie, Alberto, "Statistical nonsignificance in empirical economics," *American Economic Review: Insights*, 2020, *2* (2), 193–208.

Abbey, James D and Margaret G Meloy, "Attention by design: Using attention checks to detect inattentive respondents and improve data quality," *Journal of Operations Management*, 2017, *53*, 63–70.

Akerlof, George A., "Labor Contracts as Partial Gift Exchange," *The Quarterly Journal of Economics*, 1982, *97* (4), 543–569.

Alesina, Alberto, Stefanie Stantcheva, and Edoardo Teso, "Intergenerational Mobility and Preferences for Redistribution," *American Economic Review*, February 2018, *108* (2), 521–554.

Alvarez, R Michael and Yimeng Li, "Survey Attention and Self-Reported Political Behavior," 2021.

Anderson, Michael L., "Multiple Inference and Gender Differences in the Effects of Early Intervention: A Reevaluation of the Abecedarian, Perry Preschool, and Early Training Projects," *Journal of the American Statistical Association*, December 2008, *103* (484), 1481–1495.

Baillon, Aurelien, "Bayesian markets to elicit private information," *Proceedings of the National Academy of Sciences*, 2017.

_ , Cem Parker, and Sophie van de Zee, "Peer Betting to Elicit Unverifiable Information," 2025.

_ , Han Bleichrodt, and George Granic, "Incentives in Surveys," *Journal of Economic Psychology*, 2022.

Barseghyan, Levon, Francesca Molinari, Ted OâDonoghue, and Joshua C Teitelbaum, "Estimating risk preferences in the field," *Journal of Economic Literature*, 2018, *56* (2), 501–564.

Belloni, A., V. Chernozhukov, and C. Hansen, "Inference on Treatment Effects after Selection among High-Dimensional Controls," *The Review of Economic Studies*, April 2014, *81* (2), 608–650.

Benjamin, A. S., R. A. Bjork, and B. L. Schwartz, "The Mismeasure of Memory: When Retrieval Fluency Is Misleading as a Metamnemonic Index," *Journal of Experimental Psychology. General*, 1998.

Benjamin, Daniel, "Errors in probabilistic reasoning and judgment biases," *Handbook of Behavioral Economics: Applications and Foundations 1*, 2019.

Berinsky, Adam J., Michele F. Margolis, and Michael W. Sances, "Separating the Shirkers from the Workers? Making Sure Respondents Pay Attention on Self-Administered Surveys," *American Journal of Political Science*, July 2014, *58* (3), 739–753.

_ , _ , and _ , "Can We Turn Shirkers into Workers?," *Journal of Experimental Social Psychology*, September 2016, *66*, 20–28.

Bertrand, Marianne and Sendhil Mullainathan, "Do people mean what they say? Implications for subjective survey data," *American Economic Review*, 2001, *91* (2), 67–72.

Besken, Miri, "Generating Lies Produces Lower Memory Predictions and Higher Memory Performance than Telling the Truth: Evidence for a Metacognitive Illusion," *Journal of Experimental Psychology. Learning, Memory, and Cognition*, 2018.

Bland, J.M. and D.G. Altman, "Statistical methods for assessing agreement between two methods of clinical measurement," *Lancet*, 1986.

Chmielewski, Michael and Sarah C. Kucker, "An MTurk Crisis? Shifts in Data Quality and the Impact on Study Results," *Social Psychological and Personality Science*, May 2020, *11* (4), 464–473.

Corgnet, Brice and Roberto Hernán González, "You Will not Regret it: On the Practice of Randomized Incentives," 2023.

Curran, Paul, "Methods for the detection of carelessly invalid responses in survey data," *Journal of Experimental Social Psychology*, 2016.

Cvitanic, J., D. Prelec, B. Riley, and B. Tereick, "Honesty via Choice Matching," *American Economic Review: Insights*, 2019.

Danz, David, Lise Vesterlund, and Alistair Wilson, "Belief Elicitation and Behavioral Incentive Compatibility," *American Economic Review*, 2022.

DellaVigna, Stefano, John A List, Ulrike Malmendier, and Gautam Rao, "Estimating social preferences and gift exchange at work," *American Economic Review*, 2022, *112* (3), 1038–1074.

Dewan, Ambuj and Nathaniel Neligh, "Estimating Information Cost Functions in Models of Rational Inattention," *Journal of Economic Theory*, May 2020, *187*, 105011.

Enke, Benjamin, Uri Gneezy, Brian Hall, David Martin, Vadim Nelidov, Theo Offerman, and Jeroen Van De Ven, "Cognitive biases: Mistakes or missing stakes?," *Review of Economics and Statistics*, 2023, *105* (4), 818–832.

Falk, Armin, "Gift exchange in the field," *Econometrica*, 2007, *75* (5), 1501–1511.

_ , **Thomas Neuber, and Philipp Strack**, "Limited Self-Knowledge and Survey Response Behavior," 2021.

Fehr, Ernst and Simon Gachter, "Fairness and Retaliation: The Economics of Reciprocity," *Journal of Economic Perspectives*, 2000.

_ , **Urs Fischbacher, and Simon Gächter**, "Strong Reciprocity, Human Cooperation, and the Enforcement of Social Norms," *Human Nature*, March 2002, *13* (1), 1–25.

Frederick, Shane, "Cognitive Reflection and Decision Making," *Journal of Economic Perspectives*, 2005.

Gillen, Ben, Erik Snowberg, and Leeat Yariv, "Experimenting with Measurement Error: Techniques with Applications to the Caltech Cohort Study," *Journal of Political Economy*, 2019.

Gneezy, Uri and John A List, "Putting Behavioral Economics to Work: Testing for Gift Exchange in Labor Markets Using Field Experiments," *Econometrica*, 2006, *74* (5), 1365–1384.

Hossain, Tanjim and Ryo Okui, "The Binarize Scoring Rule," 2013.

Huang, Jason L, Paul G Curran, Jessica Keeney, Elizabeth M Poposki, and Richard P DeShon, "Detecting and deterring insufficient effort responding to surveys," *Journal of Business and Psychology*, 2012, *27*, 99–114.

Jacquemet, Nicolas, Stephane Luchini, Jason Shogren, and Verity Watson, "Discrete Choice Under Oaths," *Working paper*, 2019.

Luttmer, Erzo and Andrew Samwick, "The Welfare Cost of Perceived Policy Uncertainty: Evidence from Social Security," 2018.

Maniaci, Michael R. and Ronald D. Rogge, "Caring about Carelessness: Participant Inattention and Its Effects on Research," *Journal of Research in Personality*, February 2014, *48*, 61–83.

Mason, Winter and Duncan J Watts, "Financial incentives and the" performance of crowds"," in "Proceedings of the ACM SIGKDD workshop on human computation" 2009, pp. 77–85.

Meade, Adam W. and S. Bartholomew Craig, "Identifying Careless Responses in Survey Data.," *Psychological Methods*, September 2012, *17* (3), 437–455.

Meyer, Bruce, Wallace Mok, and James Sullivan, "Household Surveys in Crisis," 2015.

Miller, N., P. Resnick, and R. Zeckhauser, "Eliciting informative feedback: the peer prediction method," *Management Science*, 2005.

Peer, Eyal, David Rothschild, Andrew Gordon, Zak Evernden, and Ekaterina Damer, "Data quality of platforms and panels for online behavioral research," *Behavior Research Methods*, 2022.

Pew Research Center, "American Trends Panel," Survey Dataset, Pew Research Center 2024. Nationally representative panel of randomly selected U.S. adults.

Prelec, Drazen, "A Bayesian Truth Serum for Subjective Data," *Science*, 2004.

Quidt, Jonathan De, Johannes Haushofer, and Christopher Roth, "Measuring and bounding experimenter demand," *American Economic Review*, 2018, *108* (11), 3266–3302.

Radanovic, Goran and Boi Faltings, "A Robust Bayesian Truth Serum for Binary Signals," *Proceedings of the 27th AAAI Conference on Artificial Intelligence*, 2013.

Schlag, K.H., J. Tremewan, and J.J. Van Der Weele, "A Penny for Your Thoughts: A Survey of Methods for Eliciting Beliefs," *Experimental Economics*, 2015.

Schotter, Andrew and Isabel Trevino, "Belief Elicitation in the Laboratory," *Annual Review of Economics*, 2014.

Stantcheva, Stefanie, "How to run surveys: A guide to creating your own identifying variation and revealing the invisible," *Annual Review of Economics*, 2023, *15* (1), 205–234.

Vieira, Kathleen and Sean Lane, "How You Lie Affects What You Remember," *PsycEXTRA Dataset*, 2012.

Witkowski, Jens and David Parkes, "A Robust Bayesian Truth Serum for Small Populations," *Proceedings of the 26th AAAI Conference on Artificial Intelligence*, 2012.

Yammarino, Francis J, Steven J Skinner, and Terry L Childers, "Understanding mail survey response behavior a meta-analysis," *Public opinion quarterly*, 1991, *55* (4), 613–639.

Zimmermann, Florian, "The Dynamics of Motivated Beliefs," *American Economic Review*, 2020.

Online Appendix

A Supplementary tables and figures

Figure A.1: *Summary of experimental design*

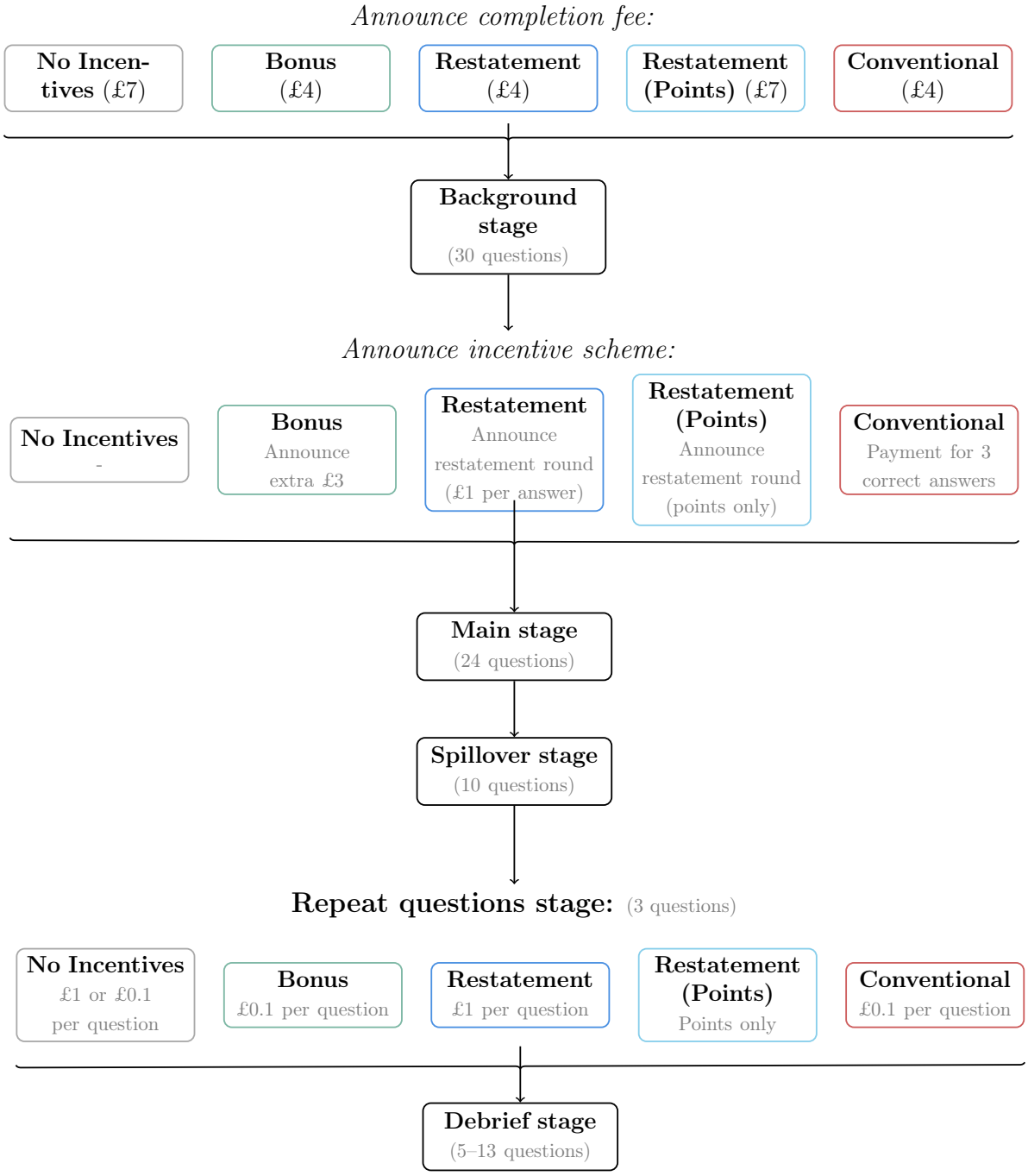
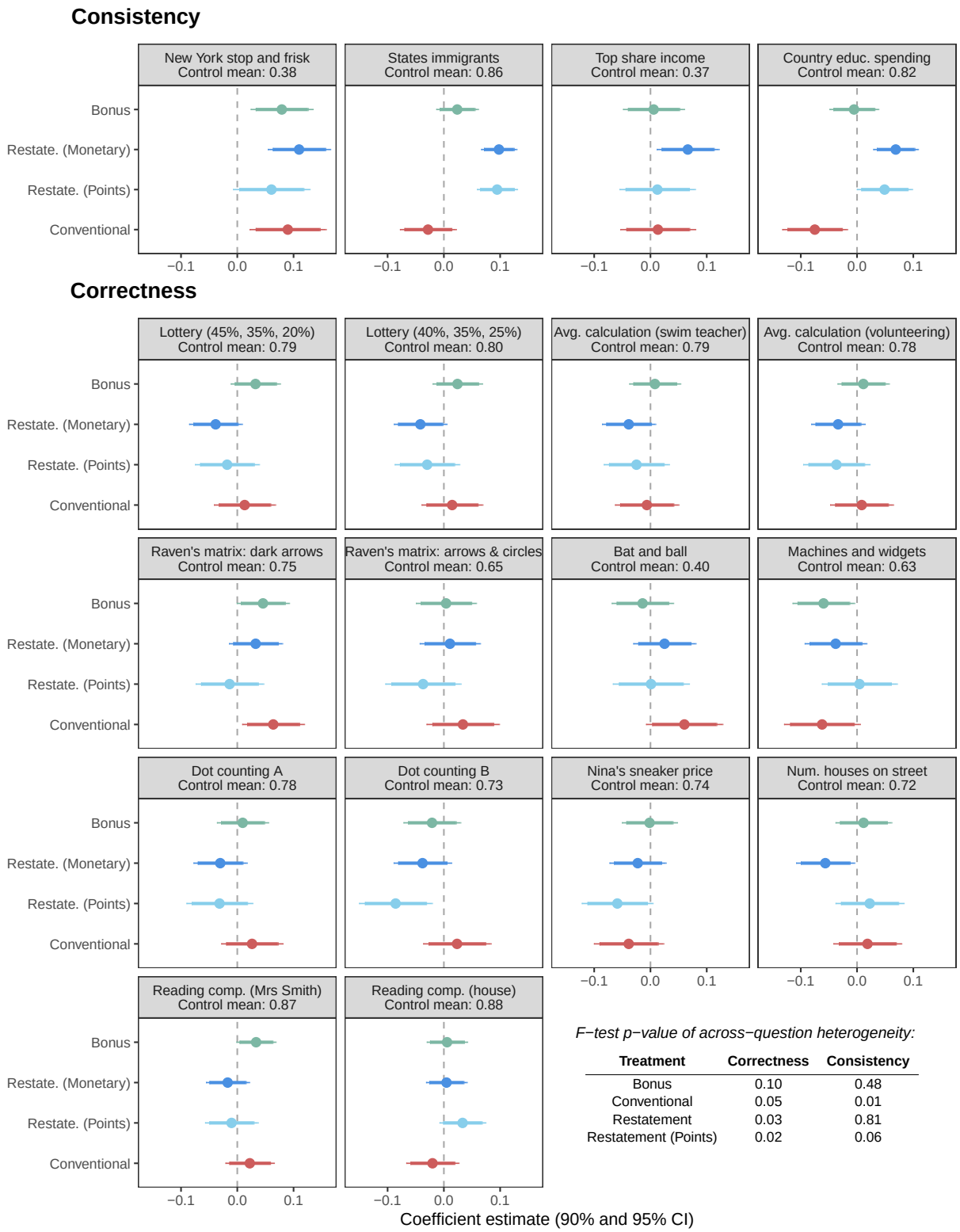


Figure A.2: *Treatment effects by question*



F−test p−value of across−question heterogeneity:

Treatment	Correctness	Consistency
Bonus	0.10	0.48
Conventional	0.05	0.01
Restatement	0.03	0.81
Restatement (Points)	0.02	0.06

Notes: F-test p-value of across-question heterogeneity is from a regression that interacts the selected treatment with all question fixed effects, and then tests the null of all interaction terms being equal to 0.

Table A.1: *Correctness questions in the main stage*

ID	Question wording
1	"A lottery ticket pays an amount of money that depends on chance. [*Probabilities are (45%, 35%, 20%), and one option first-order stochastically dominates.*] Which lottery do you most prefer?"
2	"A lottery ticket pays an amount of money that depends on chance. [*Probabilities are (40%, 35%, 25%), and one option first-order stochastically dominates.*] Which lottery do you most prefer?"
3	"William worked as a swim teacher for young kids for 5 weeks. In the first week, he worked 11 hours. In the second week, he worked 12 hours, and another 12 hours in the third week. The fourth week, he worked 8 hours. The fifth week, he worked 9 hours. How many hours did he work per week, on average?"
4	"For 5 weeks, Jennifer volunteered as a helper for swimming classes. The first week, she volunteered for 8 hours. She volunteered for 12 hours in the second week, and another 12 hours in the third week. The fourth week, she volunteered for 9 hours. The fifth week, she volunteered for 11 hours. For how many hours did she volunteer per week, on average?"
5	*Raven's matrix: dark arrows*
6	*Raven's matrix: arrows & circles*
7	"A bat and a ball cost $1.10 in total. The bat costs $1.00 more than the ball. How much does the ball cost?"
8	"If it takes 5 machines 5 minutes to make 5 widgets, how long would it take 100 machines to make 100 widgets?"
9	*Dot counting A (29 dots)*
10	*Dot counting B (31 dots)*
11	"The price of a pair of sneakers was $95 for the last six months of last year. On January 1st, there was a sale and the price decreased by 50%. 2 weeks later, the price increased by $5 and then Nina bought the sneakers at the new price. How much did Nina pay?"
12	"Dharik lives in a house on a straight street. For years, there have been 16 houses on his street to the right of his house and 17 houses on his street to the left of his house. Last year, 5 new houses were built on the same street even further to the left of those houses to the left of Dharik's house. If these are the only houses on this street, how many houses are on this street, including Dharik's house?"
13	*Reading comprehension (Mrs Smith)*
14	*Reading comprehension (house)*

Notes: All correctness questions used in the main stage. Also included were two Bayesian updating questions. These were pre-registered as exploratory, i.e., excluded from the main analysis. The questions were: (15) "The red jar contains 2 red balls and 1 blue ball. The blue jar contains 2 blue balls and 1 red ball. [*Description of 50% prior*] You reach your hand into the chosen jar, not knowing which jar it is. You have pulled out 1 blue ball. What is the probability that the red jar was chosen?" and (16) "The green jar contains 2 green balls and 1 purple ball. The purple jar contains 2 purple balls and 1 green ball. [*Description of 50% prior*] You reach your hand into the chosen jar, not knowing which jar it is. The first ball is green. The first ball is placed back in the jar. You then take another ball at random from the same jar. The second ball is also green. What is the probability that the green jar was chosen?"

Table A.2: *Consistency questions in main stage*

ID	Question wording	"Verifiable" variant
1a	"Suppose that New York City still has a 'stop and frisk' policy in the year 2035. What percentage of people being stopped in 2035 do you think will be Black or Latinx?"	Ask about actual % in 2018
1b	"Suppose that New York City still has a 'stop and frisk' policy in the year 2035. What percentage of people being stopped in 2035 do you think will be NEITHER Black NOR Latinx?"	Ask about actual % in 2018
2a	"California and Texas are sometimes thought to be the two states with the most serious problem with unauthorized immigrants. Apart from these states, which of the following states do you think has the most serious problem with unauthorized immigrants?"	Ask about number of unauthorized immigrants instead of "most serious problem"
2b	"Apart from California and Texas, which of the following US states do you think has the biggest problem with unauthorized immigrants?"	Ask about number of unauthorized immigrants instead of "most serious problem"
3a	"In the year 2050, what proportion of total income in the U.S. do you think will be earned by the 20% of people who earn the most (before tax and redistribution)?"	Ask about actual proportion in 2023
3b	"In the year 2050, what proportion of total income in the U.S. do you think will be earned by the 80% of people who earn the least (before tax and redistribution)?"	Ask about actual proportion in 2023
4a	"Which of these countries do you feel values education (primary, secondary and post-secondary non-tertiary) the most?"	Ask about country that spends most on education instead of values it the most
4b	"In terms of the value it places on education (primary, secondary and post-secondary non-tertiary), which of these countries do you feel ranks the highest?"	Ask about country that spends most on education instead of values it the most

Notes: All consistency questions used in the main stage. (1a) and (1b) are a consistency pair, whose answers should add up to 100% to be internally consistent. (3a) and (3b) must also add up to 100%. (2a) and (2b) should yield same answers; (4a) and (4b) should also yield the same answers. 50% of participants in the *Restatement*, *No Incentives*, and *Bonus* treatments were randomized to receive the "Verifiable" variant; the others received the "Unverifiable" variant.

Table A.3: *Balance for main experiment*

	Means					p-values							
	(1)	(2)	(3)	(4)	(5)								
Variable	Control	Bonus	Conventional	Restatement	Rest. (Points)	(2)–(1)		(3)–(1)		(4)–(1)		(5)–(1)	
Female (=1)	0.49	0.50	0.42	0.49	0.47	0.57		0.06	*	0.99		0.58	
Age (years)	40.20	39.38	40.71	39.73	40.52	0.30		0.60		0.57		0.75	
Married (=1)	0.33	0.28	0.35	0.31	0.28	0.09	*	0.61		0.51		0.16	
Non-White (=1)	0.23	0.29	0.30	0.28	0.30	0.01	***	0.01	**	0.04	**	0.03	**
Completed College (=1)	0.58	0.57	0.64	0.61	0.63	0.80		0.07	*	0.35		0.17	
Mother Completed College (=1)	0.44	0.45	0.43	0.44	0.45	0.81		0.84		0.97		0.78	
Father Completed College (=1)	0.43	0.41	0.44	0.45	0.45	0.58		0.68		0.47		0.54	
Employed (=1)	0.66	0.66	0.68	0.65	0.62	0.93		0.56		0.74		0.26	
Passed Attention Check (=1)	0.90	0.86	0.91	0.89	0.90	0.03	**	0.55		0.64		0.88	
Born in USA (=1)	0.95	0.94	0.94	0.97	0.93	0.32		0.62		0.16		0.19	
Household Size (#)	2.68	2.76	2.70	2.66	2.74	0.33		0.85		0.78		0.54	
Children in Household (# under 18)	0.42	0.53	0.51	0.46	0.49	0.03	**	0.18		0.44		0.31	
Household Income ($1000s/year)	67.25	61.06	65.95	61.61	64.14	0.01	**	0.65		0.02	**	0.32	
Religious (=1)	0.43	0.43	0.45	0.41	0.44	0.90		0.61		0.42		0.67	
Homeowner (=1)	0.49	0.45	0.52	0.44	0.46	0.21		0.39		0.13		0.49	
Registered to Vote (=1)	0.92	0.90	0.89	0.88	0.90	0.34		0.26		0.07	*	0.53	
Altruism Index (Z-score)	0.00	-0.09	-0.01	-0.07	-0.05	0.11		0.89		0.27		0.52	
F-test: statistic						**1.63**		**1.12**		**1.06**		**0.63**	
F-test: p-value						**0.05**		**0.32**		**0.39**		**0.87**	

Notes: Columns (1)-(5) show the means of the demographic variables in each treatment group. Columns 6-9 show the p-value of the difference between each treatment group and the *No Incentives* control group. F-test is the test of the joint null of $\beta_1 = \beta_2 = ... = \beta_K = 0$ for the regression $Treat_i = \alpha + \sum_{k=1}^{K} X_i^k + \varepsilon_i$, where X_i^k are the demographic variables, $Treat_i$ is whether the individual is in the relevant treatment group or in the *No Incentives* control group.

Table A.4: *Attrition and selection*

Outcome var:	Completed survey (=1)		Predicted correctness			Predicted consistency		
Sample:	Started survey	Completed demographics	Completed demographics	Completed survey	Completed survey (adjusting for attrition)	Completed demographics	Completed survey	Completed survey (adjusting for attrition)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Bonus	0.005	0.006	−0.013**	−0.014**	−0.014**	−0.009*	−0.010*	−0.010*
	(0.009)	(0.006)	(0.006)	(0.006)	(0.006)	(0.005)	(0.005)	(0.005)
Conventional	0.005	0.005	−0.003	−0.003	−0.003	−0.007	−0.007	−0.007
	(0.011)	(0.007)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)
Restatement	−0.016	0.001	−0.003	−0.004	−0.004	−0.008*	−0.009*	−0.009*
	(0.011)	(0.007)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)
Restatement-Points	−0.017	−0.015	0.002	0.000	0.000	−0.006	−0.007	−0.007
	(0.014)	(0.011)	(0.006)	(0.006)	(0.007)	(0.006)	(0.006)	(0.006)
Num. participants	2507	2462	2462	2428	2428	2462	2428	2428
R-squared	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.01
Dep. var. control mean	0.97	0.99	0.75	0.75	0.75	0.60	0.60	0.60

Notes: Outcome in columns 1-2 is a binary indicator of whether the participant completed the survey. Column 1 includes all individuals who started the survey; column 2 includes all individuals who completed the baseline demographic survey. Outcome in columns 3-5 is *Predicted correctness*. This is constructed by predicting the correctness of each individual participant based on their baseline demographics (variables in Table A.3), using coefficients from an OLS regression of the baseline demographics on correctness score in the *No Incentives* group. Outcome in columns 6-8 is *Predicted consistency*, which is constructed similarly. Columns 3 and 6 include all participants who completed the demographic survey; columns 4 and 7 include all participants who completed the survey (and therefore are included in the main specification). Columns 5 and 8 adjust for within-survey attrition by predicting the probability of attrition using a logistic regression with baseline demographics and reweighting the sample using inverse-probability weights. Standard errors in columns 3-8 are bootstrapped with 500 repetitions to account for error in the calculation of the inverse-probability weights and the predicted correctness and consistency scores.

Table A.5: *Impacts are robust to multiple hypothesis testing and alternative specifications*

	Correctness				Consistency			
	Correct Answer (=1) (MHT q-values)	Correct Answer (=1) (RI p-values)	Correctness, Anderson index (Z)	Correct Answer (=1) Drop failed attention checks	Consistent Answer (=1) (MHT q-values)	Consistent Answer (=1) (RI p-values)	Consistency, Anderson index (Z)	Consistent Answer (=1) Drop failed attention checks
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Bonus	0.021**	0.021**	0.116**	0.019*	0.030**	0.030**	0.149***	0.032**
	(0.010) [0.050]	(0.010) [0.032]	(0.050) [0.019]	(0.011) [0.076]	(0.013) [0.050]	(0.013) [0.014]	(0.051) [0.003]	(0.014) [0.021]
Restatement	−0.014	−0.014	−0.043	−0.019	0.084***	0.084***	0.391***	0.087***
	(0.011) [0.150]	(0.011) [0.192]	(0.050) [0.393]	(0.012) [0.106]	(0.013) [<0.001]	(0.013) [<0.001]	(0.051) [<0.001]	(0.014) [<0.001]
Restatement-Points	−0.019	−0.019	−0.081	−0.013	0.070***	0.070***	0.335***	0.070***
	(0.013) [0.140]	(0.013) [0.126]	(0.065) [0.214]	(0.015) [0.395]	(0.017) [<0.001]	(0.017) [<0.001]	(0.067) [<0.001]	(0.018) [<0.001]
Conventional	0.015	0.015	0.074	0.015	0.012	0.012	0.030	0.013
	(0.013) [0.170]	(0.013) [0.232]	(0.065) [0.255]	(0.014) [0.279]	(0.018) [0.310]	(0.018) [0.512]	(0.067) [0.652]	(0.019) [0.503]
Num. observations	33 992	33 992	2428	30 212	9712	9712	2428	8632
Num. participants	2428	2428	2428	2158	2428	2428	2428	2158
Dep. var. control mean	0.74	0.74	0.00	0.74	0.61	0.61	0.00	0.61
R-squared	0.14	0.14	0.28	0.12	0.25	0.25	0.12	0.25
Question FEs	X	X		X	X	X		X
LASSO controls	X	X	X	X	X	X	X	X
p-val: Bonus=Conv.	0.61	0.49	0.56	0.79	0.31	0.19	0.09*	0.31
p-val: Restate.=Bonus	0.00***	0.01**	0.00***	0.00***	0.00***	0.00***	0.00***	0.00***
p-val: Restate.=Conv.	0.03**	0.00***	0.07*	0.02**	0.00***	0.00***	0.00***	0.00***
p-val: Restate.=Restate-Points	0.69	0.61	0.59	0.70	0.38	0.37	0.48	0.32
p-val: Restate. (Mon. + Points pooled)=0	0.11	0.04**	0.19	0.11	0.00***	0.00***	0.00***	0.00***

Notes: Column 1 (correctness) and 5 (consistency) use q-values adjusted for multiple hypothesis testing to control the false discovery rate (Anderson, 2008); standard errors in parentheses, q-values in brackets; * $q<0.1$, ** $q<0.05$, *** $q<0.01$. Columns 2 and 6 use randomization inference p-values based on 500 simulations. Columns 3 and 7 use inverse-covariance-weighted indexes at the participant level (Anderson, 2008). Columns 4 and 8 drop participants who failed the attention check. Specification is otherwise the same as Table 1.

Table A.6: *Heterogeneity by question difficulty and baseline response time*

	By difficulty	By baseline response time			
	Correct (=1)	Consistent Answer (=1)		Correct Answer (=1)	
	(1)	(2)	(3)	(4)	(5)
Bonus	0.002	0.030**	0.014	0.022**	0.018
	(0.013)	(0.013)	(0.019)	(0.010)	(0.014)
Conventional	0.009	0.012	0.024	0.015	0.027
	(0.017)	(0.018)	(0.026)	(0.013)	(0.018)
Restatement	−0.014	0.084***	0.086***	−0.013	−0.014
	(0.014)	(0.013)	(0.018)	(0.011)	(0.015)
Restatement-Points	−0.025	0.070***	0.073***	−0.019	−0.012
	(0.017)	(0.017)	(0.023)	(0.013)	(0.019)
Above med. control mean (=1)	0.163***				
	(0.009)				
Bonus × Above med. control mean (=1)	0.035***				
	(0.013)				
Conventional × Above med. control mean (=1)	0.009				
	(0.016)				
Restatement × Above med. control mean (=1)	0.000				
	(0.013)				
Restatement-Points × Above med. control mean (=1)	0.009				
	(0.016)				
Above med. baseline response time		−0.004	−0.008	0.017**	0.019
		(0.010)	(0.019)	(0.008)	(0.014)
Bonus × Above med. baseline response time			0.033		0.009
			(0.026)		(0.020)
Conventional × Above med. baseline response time			−0.022		−0.023
			(0.034)		(0.025)
Restatement × Above med. baseline response time			−0.004		0.001
			(0.026)		(0.021)
Restatement-Points × Above med. baseline response time			−0.005		−0.013
			(0.032)		(0.025)
Num. observations	33 992	9712	9712	33 992	33 992
Num. participants	2428	2428	2428	2428	2428
Dep. var. control mean	0.74	0.61	0.61	0.74	0.74
R-squared	0.11	0.25	0.25	0.14	0.14
Question FEs		X	X	X	X

Table A.7: *Consistency is slightly lower for verifiable questions, and is unaffected by question positioning*

	Consistent Answer (=1)			
	(1)	(2)	(3)	(4)
Bonus	0.030**	0.024	0.030**	0.030**
	(0.013)	(0.018)	(0.013)	(0.013)
Conventional	0.012	0.014	0.012	0.012
	(0.018)	(0.020)	(0.018)	(0.018)
Restatement	0.084***	0.084***	0.084***	0.084***
	(0.013)	(0.018)	(0.013)	(0.013)
Restatement-Points	0.070***	0.072***	0.070***	0.070***
	(0.017)	(0.018)	(0.017)	(0.017)
Verifiable questions (=1)	−0.021*	−0.025		
	(0.011)	(0.019)		
Bonus × Verifiable questions (=1)		0.012		
		(0.026)		
Restatement × Verifiable questions (=1)		0.001		
		(0.026)		
Position of pair (Z)			−0.008	
			(0.008)	
Bonus × Position of pair (Z)			0.015	
			(0.012)	
Conventional × Position of pair (Z)			0.003	
			(0.015)	
Restatement × Position of pair (Z)			0.002	
			(0.012)	
Restatement-Points × Position of pair (Z)			0.018	
			(0.014)	
Question distance (Z)				0.005
				(0.009)
Bonus × Question distance (Z)				0.002
				(0.012)
Conventional × Question distance (Z)				0.004
				(0.016)
Restatement × Question distance (Z)				−0.012
				(0.012)
Restatement-Points × Question distance (Z)				0.004
				(0.015)
Num. observations	9712	9712	9712	9712
Num. participants	2428	2428	2428	2428
Dep. var. control mean	0.61	0.61	0.61	0.61
R-squared	0.25	0.25	0.25	0.25
Question FEs	X	X	X	X

Conventional and *Restatement-Points* arms only used verifiable questions, so the interaction terms are excluded in column 2. *Position of pair (Z)* is order of appearance of the questions in the consistency pair, averaged over the two questions and then standardized to have a mean of 0 and standard deviation of 1. *Question distance (Z)* is a measure of the number of questions *between* the two questions in the consistency pair, standardized in the same way. Specification is otherwise the same as Table 1, but without LASSO controls.

Table A.8: *Comparison to attention check*

	Full Sample (1)	Passed Attention Check (2)	Effect Drop Failed Att. Check (2) - (1)	Bonus Effect	p-val. Bonus vs Drop
Panel A: All treatments					
Correctness	0.733	0.744	0.011	0.021	0.29
Consistency	0.643	0.650	0.007	0.030	0.10
N	2,428	2,158	-270 (11.1%)		
Panel B: No Incentives only					
Correctness	0.737	0.745	0.008	0.021	0.19
Consistency	0.608	0.614	0.005	0.030	0.06
N	609	548	-61 (10.0%)		

Notes: Shows the proportion of correct and consistent answers for the full sample (column 1) and when dropping people who fail the attention check (column 2). *Column 3*: the effect of dropping people who fail the attention check. *Column 4*: effect of *Bonus* from Table 1. *Column 5*: p-values comparing the effect of *Bonus* and the effect of dropping participants who failed the attention check. Null distribution to calculate p-values is calculated by randomly permuting the *No Incentives* and *Bonus* 1000 times, randomly permuting who failed attention checks 1000 times (keeping proportions fixed), and comparing the 2 values. *Panel A* includes all treatment groups for columns (1)-(3), *Panel B* includes only *No Incentives* participants. In Panel A, participants who *pass* the attention check are 6.4 pp (p<0.001) more correct and 4.8 pp (p=0.001) more consistent than those who *fail* the attention check.

Table A.9: *Correlation questions in the supplementary experiment*

ID	Pew ID	Question subject	ATP Version	Correlation in Pew
1a	policy3	Scientific experts good at policy decisions	114	-0.46
1b	policy1b	Scientists have too much/too little influence		
2a	abrtlgl	Do you think abortion should be legal	110	-0.68
2b	abrtn_scotus	Do you approve of repeal of Roe v Wade		
3a	satisfied_democracy	Satisfied with democracy	105	0.34
3b	efficacy	Political system gives people like me a voice		
4a	intthreat_climatechange	How much of a threat, if at all, is climate change to our country?	105	0.73
4b	citizen_environment	How important is helping reduce climate change		
5a	smimpact_informed	Social media - made people more or less informed	105	0.31
5b	smeffect_awareness	Social media - effective for public awareness		
6a	fav_nato	Favorable about NATO	105	0.54
6b	benefitorg_nato	US benefits from NATO		
7a	transgend7	Discrimination against trans	109	0.46
7b	transgend6_a	Support protection against trans discrimination		
8a	techinfl	Tech have too much power/influence	88	0.38
8b	tc5	Tech should be regulated		
9a	ewtfreq	Extreme weather more than in the past	95	0.34
9b	infrgoal_d	Gov should set stricter standards for extreme weather		
10a	natprobs_k	Climate change is a problem	53	0.69
10b	clim1a	Climate is getting warmer because of people		
11a	natprobs_j	Economic inequality is a problem	53	-0.27
11b	taxsysm	Federal tax system is fair		
12a	natprobs_i	Illegal immigration is problem	53	0.65
12b	immva_lb	Should increase deportations		

Notes: All correlation questions used in the supplementary experiment. All questions taken from Pew's American Trends Panel Survey (Pew Research Center, 2024). *Pew ID* indicates the ID of the question in the original Pew dataset; ATP version indicates the wave of the ATP survey it was contained in. *Correlation in Pew* indicates the Pearson correlation between the two questions in the pair in the original Pew dataset, based on authors' own calculations. All questions had ordered lists as responses (e.g. (3a): "How satisfied are you with the way democracy is working in the U.S.?", options: *Very satisfied*, *Somewhat satisfied*, *Not too satisfied*, *Not at all satisfied*). These were coded as integers to calculate correlations. For pairs with *negative* correlations, one item was reverse coded in the supplementary experiment so that all correlations were in the same direction in the experimental data.

Table A.10: *Supplementary experiment: impact on correctness and correlations.*

	Correct Answer (=1)		Correlation pair: Answer 1	
	(1)	(2)	(3)	(4)
Restatement-Points	−0.012	−0.007	−0.005	−0.004
	(0.012)	(0.011)	(0.015)	(0.014)
Answer 2			0.471***	0.463***
			(0.013)	(0.012)
Answer 2 × Restatement-Points			−0.026	−0.025
			(0.018)	(0.018)
Num. observations	19 684	19 684	16 872	16 872
Num. participants	1408	1408	1408	1408
Dep. var. control mean	0.62	0.62	0.00	0.00
R-squared	0.11	0.15	0.22	0.23
Question FEs	X	X		
Question-pair FEs			X	X
LASSO controls		X		X

Notes: Data from the supplementary experiment. Columns 1-2 use the same specification as Table 1. Columns 3-4 specification is Equation 11. *Answer 1* is the first question in a pair in the survey, *Answer 2* is the second question in a pair in the survey. All correlation questions have ordered lists as responses, which are coded as integers, then standardized so that the mean is 0 and standard deviation is 1 in the *No Incentives* (control) group. ρ, the average correlation between the two options in a pair, is thus 0.46. Standard errors clustered at the participant level. Omitted category is the *No Incentives* arm.

Table A.11: *Results are similar when dropping participants who report gaming*

	Drop gamers		Effect of gaming	
	Correct (=1)	Consistent (=1)	Correct (=1)	Consistent (=1)
	(1)	(2)	(3)	(4)
Bonus	0.022**	0.031**	0.021**	0.030**
	(0.010)	(0.013)	(0.010)	(0.013)
Conventional	0.015	0.012	0.015	0.012
	(0.013)	(0.018)	(0.013)	(0.018)
Restatement	−0.009	0.081***	−0.010	0.081***
	(0.011)	(0.014)	(0.011)	(0.014)
Restatement-Points	−0.018	0.068***	−0.016	0.068***
	(0.014)	(0.018)	(0.014)	(0.017)
Gamer (Restate. arms only)			−0.019	0.014
			(0.017)	(0.018)
Num. observations	31 514	9004	33 992	9712
Num. participants	2251	2251	2428	2428
Dep. var. control mean	0.74	0.61	0.74	0.61
R-squared	0.14	0.25	0.14	0.25
Question FEs	X	X	X	X
LASSO controls	X	X	X	X
p-val: Bonus=Conv.	0.60	0.31	0.60	0.31
p-val: Restate.=Bonus	0.01***	0.00***	0.01***	0.00***
p-val: Restate.=Conv.	0.08*	0.00***	0.07*	0.00***
p-val: Restate.=Restate-Points	0.55	0.48	0.64	0.41
p-val: Restate. (Mon. + Points pooled)=0	0.23	0.00***	0.23	0.00***

Notes: Columns 1 and 2 drop all 177 participants in *Restatement* or *Restatement-Points* who report using any gaming strategy (choosing easy-to-remember answers, searching for answers online, writing answers down). Columns 3 and 4 retain these participants but include a binary indicator for gamers as an explanatory variable.

B Conceptual framework

A conceptual framework illustrates how the Bonus and Restatement methods may affect the accuracy of survey responses. Suppose the respondent has the following utility function:

$$U\left(e_1, e_2; B, R\right) = \underbrace{\gamma(B) p_Q\left(e_1, e_2\right)}_{\text{intrinsic payoff for accuracy}} + \underbrace{R p_R\left(e_1, e_2\right)}_{\text{restatement payoff}} - \underbrace{C\left(e_1, e_2\right)}_{\text{effort cost}}, \tag{2}$$

where $p_Q(e_1, e_2)$ is the probability of answering a given question accurately, which depends on two dimensions of effort: providing careful initial responses (e_1) and memorizing answers or gaming (e_2); $\gamma(B) \geq 0$ is the intrinsic utility from answering accurately, which may depend on the bonus payment B that is paid regardless of accuracy;[29] $p_R(e_1, e_2)$ is the probability of correctly restating the original answers on repeat questions Q_R, which also depends on both dimensions of effort; R is the restatement payment; and $C(e_1, e_2)$ is the cost of effort. $B > 0$ and $R = 0$ corresponds to the *Bonus* treatment, $B = 0$ and $R > 0$ corresponds to the *Restatement* treatment, and $B = R = 0$ corresponds to the *No Incentives* treatment.

For simplicity, we assume linear functional forms for $p_Q(\cdot)$ and $p_R(\cdot)$. Answering carefully improves initial accuracy, but memorizing or gaming may hurt accuracy, so $p_Q(e_1, e_2) = p_0 + \alpha_1 e_1 - \alpha_2 e_2$ with $\alpha_1 > 0$, $\alpha_2 \geq 0$. By contrast, answering carefully, memorizing, and gaming all increase the probability of correctly restating, so $p_R(e_1, e_2) = p_0' + \beta_1 e_1 + \beta_2 e_2$ with $\beta_1 > 0, \beta_2 > 0$. We assume that costs are separable and convex, i.e., $C(e_1, e_2) = c_1(e_1) + c_2(e_2)$ with $c_j(0) = 0$, $c_j'(e_j) > 0$, and $c_j''(e_j) > 0$ for $e_j > 0$ and $j \in \{1, 2\}$.[30]

Bonus treatment. When $R = 0$, the participant will set $e_2 = 0$ and e_1 such that $\gamma(B)\alpha_1 = c_1'(e_1)$. Increasing the bonus payment will therefore increase effort e_1 and initial

[29] We allow for $\gamma(0) > 0$, so there can be an intrinsic desire to answer accurately even without any bonus payment.

[30] We also assume two technical conditions: that each $c_j(\cdot)$ and $\gamma(\cdot)$ are twice continuously differentiable, so that the previous conditions are well-defined; and that $C(\cdot)$ and the other parameters ensure that the optimal effort choices are strictly interior as long as $\gamma(B) > 0$ and $R > 0$.

answer accuracy p_Q if and only if $\gamma'(B) > 0$. In other words, if the gift increases the intrinsic payoff for accuracy, then accuracy will increase.

Restatement treatment. We also derive the conditions under which increasing the restatement incentive R can increase initial accuracy p_Q.

Proposition 1.

$$\frac{\partial p_Q^*}{\partial R} > 0 \quad \textit{if and only if} \quad \frac{\alpha_1 \beta_1}{c_1''(e_1^*(R))} > \frac{\alpha_2 \beta_2}{c_2''(e_2^*(R))} \tag{3}$$

Proof of Proposition 1. Consider a participant who chooses effort levels $(e_1 \geq 0, e_2 \geq 0)$ to maximize the utility

$$U\big(e_1, e_2; B, R\big) \;=\; \underbrace{\gamma(B)\, p_Q\big(e_1, e_2\big)}_{\text{intrinsic payoff for accuracy}} \;+\; \underbrace{R\, p_R\big(e_1, e_2\big)}_{\text{restatement payoff}} \;-\; \underbrace{C\big(e_1, e_2\big)}_{\text{effort cost}}, \tag{4}$$

First-Order Conditions. For a given B and R, the participant chooses (e_1, e_2) to maximize (4). The first-order conditions (FOCs) for an interior solution (e_1^*, e_2^*) are

$$\frac{\partial U}{\partial e_1} = \gamma(B)\, \alpha_1 \;+\; R\, \beta_1 \;-\; c_1'\big(e_1\big) \;=\; 0, \tag{5}$$

$$\frac{\partial U}{\partial e_2} = \gamma(B)\, \big(-\alpha_2\big) \;+\; R\, \beta_2 \;-\; c_2'\big(e_2\big) \;=\; 0. \tag{6}$$

Hence, at the optimum:

$$c_1'\big(e_1^*\big) = \gamma(B)\, \alpha_1 \;+\; R\, \beta_1, \tag{7}$$

$$c_2'\big(e_2^*\big) = -\,\gamma(B)\, \alpha_2 \;+\; R\, \beta_2. \tag{8}$$

Because c_1' and c_2' are strictly increasing (each c_i is strictly convex), these implicitly define unique effort choices $e_1^*(R)$, $e_2^*(R)$ for any given B and R. Inverse functions of the form

$e_1^*(R) = \left[c_1'\right]^{-1}\big(\gamma(B)\,\alpha_1 + R\,\beta_1\big)$, $e_2^*(R) \;=\; \left[c_2'\right]^{-1}\big(-\,\gamma(B)\,\alpha_2 + R\,\beta_2\big)$ obtain if c_i' is invertible.

Condition for $\dfrac{\partial p_Q^*}{\partial R} > 0$. We define *initial accuracy*, or the probability of answering question Q correctly in the first stage, as $p_Q^*(R) \;\equiv\; p_Q\big(e_1^*(R),\, e_2^*(R)\big) \;=\; p_0 \;+\; \alpha_1\, e_1^*(R) \;-\; \alpha_2\, e_2^*(R)$. To study how $p_Q^*(R)$ changes with the restatement incentive R, take the total derivative:

$$\frac{d}{dR}\, p_Q^*(R) \;=\; \alpha_1\, \frac{\partial e_1^*}{\partial R} \;-\; \alpha_2\, \frac{\partial e_2^*}{\partial R}.$$

From (7) and (8), we have

$$\frac{\partial e_1^*}{\partial R} = \frac{\beta_1}{c_1''\big(e_1^*\big)}, \quad \frac{\partial e_2^*}{\partial R} = \frac{\beta_2}{c_2''\big(e_2^*\big)},$$

assuming differentiability of c_i' and $c_i'' > 0$. Therefore,

$$\frac{d}{dR}\, p_Q^*(R) = \alpha_1\, \frac{\beta_1}{c_1''\big(e_1^*\big)} \;-\; \alpha_2\, \frac{\beta_2}{c_2''\big(e_2^*\big)}. \tag{9}$$

Equation (9) reveals that raising R *increases* initial (Stage-1) accuracy p_Q^* if and only if

$$\alpha_1\, \frac{\beta_1}{c_1''\big(e_1^*\big)} \;>\; \alpha_2\, \frac{\beta_2}{c_2''\big(e_2^*\big)}. \tag{10}$$

□

The restatement payment R will thus improve accuracy if (i) answering carefully contributes strongly to answering accurately (high α_1); (ii) answering carefully strongly improves one's ability to restate in the next round (high β_1); and (iii) exerting more effort on e_1 is relatively "cheap" (low $c_1''(.)$). Conversely, increasing R will worsen accuracy if (iv) memorizing or gaming strongly worsens accuracy (high α_2); (v) memorizing or gaming is effective at improving restatement (high β_2); and (vi) memorizing or gaming is "cheap" (low $c_2''(.)$).

C Supplementary experiment description

We ran a supplementary experiment (October–December 2024) to determine whether the restatement method improves answer quality when participants cannot rely on recognizing near-duplicates. The design mirrors our main experiment with specific modifications outlined below.

C.1 Supplementary experiment: design

Treatments. Given the evidence that *Restatement-Points* was effective at increasing consistency in the main experiment, we include two treatments: *Restatement-Points* (N=702) and *No Incentives* (N=706). Both groups are paid a £4 completion fee, and are not paid to correctly restate answers in the repeat questions stage, yielding a clean comparison in which total payments are always exactly the same. The instructions are otherwise the same as in the main experiment.

Stages. The stage structure was the same as in the main experiment, except the spillover stage was removed.

Main stage questions and outcomes.

Correctness. To replicate the results on correctness without any concern about near-duplicate questions, we included 14 correctness questions. We started from the same set as the main experiment, but replaced questions that could be perceived as duplicates (e.g., one of the two dot counting tasks) with additional Raven's matrices.[31]

Correlation. To measure answer quality on unverifiable questions that were *not* near-duplicates, we included 12 pairs of questions from Pew's American Trends Panel Survey (ATP) (Pew Research Center, 2024) that are highly correlated in the Pew data (average $|\rho|$=0.49), and are a priori strongly related (see Table A.9 for question details). In each pair,

[31]Specifically, we removed questions 2, 4, 10, and 14 from Table A.1.

one question concerns a *belief* related to a policy issue, and another concerns a corresponding *policy preference*.[32] If *Restatement-Points* increased answer quality, we would expect to see less noise in the data and thus higher correlations within pairs.

Empirical strategy for correlation. For each pair, we call the first question in the survey *Answer 1*, and the second question *Answer 2*. We standardize both so that the standard deviation in the *No Incentives* group is 1.[33] Then, the specification for participant i and question-pair j is:

$$Answer1_{ij} = \alpha Answer2_{ij} + \beta RestatePoints_i + \delta(Answer2_{ij} \times RestatePoints_i) + \varepsilon_{ij} \tag{11}$$

where α is the average correlation between the two answers in the *No Incentives* group and δ is the main coefficient of interest that indicates whether the correlation changes in the *Restatement-Points* arm.

Results. The *Restatement-Points* treatment does not increase the correlation between these pairs (β=-0.025, p=0.16, Table A.10, columns 3-4), despite our large sample size (N=1408) providing sufficient power to detect even small changes in correlation of 0.05.[34] We also use the supplementary experiment to replicate the results on correctness without potentially distracting near-duplicate questions, and again find no effect on correctness (p=0.48, Table A.10, columns 1-2). Together, these results show that *Restatement*-based methods enhance internal consistency of answers on near-duplicate questions, but have no detectable impacts on other measures of answer quality.

[32]For example, we asked participants (3a) "How satisfied are you with the way democracy is working in the U.S.?" (Options: *Very satisfied, Somewhat satisfied, Not too satisfied, Not at all satisfied*) and (3b) "How much would you say the political system in the U.S. allows people like you to have an influence on politics?" (Options: *A great deal, A fair amount, Not too much, Not at all*).

[33]All answer options are either ordered multiple choice (which we treat as integers) or binary. For pairs with negative correlations, we reverse-code one of the questions in the pair so that all correlations are expected to be positive in our data.

[34]This assumes α=5%, β=80%, implying that minimum detectable effect is $2.8\hat{\sigma}$ where $\hat{\sigma}$ is the standard error of the estimate.

D Data and measurement

D.1 Recall strategy coding and prompt

Model: GPT-4o-mini, temperature set at 0.

```
You are an AI assistant helping researchers analyze responses from an online experiment. Your
    task is to categorize participants' strategies based on their responses to a question about
     how knowing they would have to restate their answers later affected their initial
    responses.

Here is the participant's response:
<participant_response>
{{RESPONSE INSERTED HERE}}
</participant_response>

Your goal is to determine whether the participant used any of the following strategies:
1. Extra attention: They answered with extra attention, effort, or truthfulness in their
    initial answers, taking care to provide accurate responses in the first place so they could
     restate more easily later.
2. Memorize: They tried to memorize their answers, so they could restate them later more easily
3. Game the system: They tried to game the system (e.g., by writing down their answers,
    searching online, using AI, or other methods).
4. No strategy: They had no particular strategy.
5. Other: They used a strategy not covered by the above categories.

Please follow these steps:
1. Carefully read and analyze the participant's response.
2. In <strategy_analysis> tags, for each strategy:
   a. Provide arguments for why the participant might have used this strategy.
   b. Provide arguments for why the participant might not have used this strategy.
   c. Make a final decision on whether the participant used this strategy, based on the
       arguments.
3. Based on your analysis, create a JSON object with the following keys: extra_attention,
    memorize, game, no_strategy, and other. Assign a value of 1 to each strategy the
    participant used and 0 to those they didn't use.

Example output structure (do not copy the content, only the structure):
<strategy_analysis>
Extra attention:
Arguments for: [Reasoning]
Arguments against: [Reasoning]
Decision: [Used/Not used]

Memorize:
[Same structure as above]
```

```
Game the system:
 [Same structure as above]
No strategy:
 [Same structure as above]
Other:
 [Same structure as above]
</strategy_analysis>

{
  "extra_attention": 0,
  "memorize": 0,
  "game": 0,
  "no_strategy": 0,
  "other": 0
}
```

D.2 LASSO controls

Following Belloni et al. (2014), we use double LASSO to select controls in the main results. The variables selected for the main specifications were:

Consistency (set $\mathcal{C}_1$): *Widowed, Lives in Puerto Rico, Lives in South Dakota, # home purchases in last 12 months, Purchased Amazon Prime membership, Purchased books, Gets news from smart speaker, Gets news from smartwatch, Confidence in religious leaders, Confidence in business leaders, Response time, Verifiable or unverifiable.*

Correctness (set $\mathcal{C}_2$): selected variables include all those included in the consistency measures, plus: *African-American, Asian-American, High school degree, Atheist, Passed baseline attention check, # home appliances owned,*
 # modes of transportation used, Owns dishwasher, Owns smoke alarm, Owns vacuum cleaner, Walking for commute, Purchased spectator sports tickets, Purchased DVDs or video streaming), Got news from NYT, Speaks Mandarin fluently, Believes ABC News does own reporting, Online access makes gov. more accessible, Response time.

For other analyses, we control for (i) $\mathcal{C}_1$ when the outcome is consistency; (ii) $\mathcal{C}_2$ when the outcome is correctness; and (iii) the union of both sets ($\mathcal{C}_1 \cup \mathcal{C}_2$) for other outcomes.